Appendix Table 1. General characteristics of studies included in the systematic review

Study ID	Region	Period of Data Collection	Centers (n)	Sample size for analysis (n)	Event	Study design	Data sources	Age (Mean±SD or median (interquartile range)) (years)	Male (%)	Study purpose
Liu et al (2018a)	China	2006.01-2017.01	1	143	32	Retrospective cohort	EMR data	50.0 (43.0, 62.0)	72.00%	Prediction performance of prognostic factors
Zindovic et al (2018)	Sweden	2005.01-2017.02	1	277	37	Retrospective cohort	EMR data	63.3±11.4	63.86%	Prediction performance of prognostic factors
Oz et al (2017)	Turkey		1	57	15	Retrospective cohort	EMR data	54.6±10.5	15.80%	Prediction performance of prognostic factors
Li et al (2016)	China	2010.05-2014.06	4	103	36	Prospective cohort	EMR data	54.5±13.4	68.93%	Prediction performance of prognostic factors
Vrsalovic et al (2015)	Croatia	2006.01-2013.12	1	54	24	Retrospective cohort	EMR data	69.0±14.0	63.00%	External validation
Karakoyun et al (2015)	Turkey	2009-2013	1	35	9	Retrospective cohort	EMR data	55.91±7.95	80.00%	Prediction performance of prognostic factors
Wen et al (2019)	China	2008.03-2012.01	1	122	29	Prospective cohort	Cohort	50.04±8.35	84.43%	Prediction performance of prognostic factors
Liu et al (2018b)	China	2012.12-2016.06	1	192	19	Retrospective cohort	EMR data	51.0 (44.0, 62.0)	78.60%	Prediction performance of prognostic factors
Bennett et al (2017)	USA	2000-2014	1	144	38	Retrospective cohort	EMR data	58.7 (48.9, 69.7)	67.00%	Prediction performance of prognostic factors
Zhang et al (2015)	China	2008.01-2013.10	1	360	77	Prospective cohort	Cohort	57.1±12.6	75.80%	Develop a model without validation

LAFÇI et al (2014)	Turkey	2007.01-2012.01	1	104	33		Retrospective cohort	EMR data	55.2±14.0	73.08%	Prediction performance of prognostic factors
Wen et al (2013)	China	2007.01-2011.10	1	114	31		Prospective cohort	Cohort	48.8±7.6	84.20%	Prediction performance of prognostic factors
Guo et al (2019)	China	2015.12-2017.08	1	109	31		Prospective cohort	Cohort	52.0±12.3	59.63%	Prediction performance of prognostic factors
Ohlmann et al (2006)	France	1997.01-2003.12	1	93	22		Retrospective cohort	EMR data	63.8±12.6	66.00%	Prediction performance of prognostic factors
Ge et al (2013)	China	2009.02-2012.02	1	384	31		Retrospective cohort	Cohort	46.0±10.8	20.05%	External Validation
Tolenaar et al (2014)	Multinational	1996.01-2013.04	Multicenter	1034	110		Prospective cohort	Registry	63.5±14.0	65.10%	Develop a model without validation
Mehta et al (2002)	6 countries	1996.01-1999.12	18	547	178		Prospective cohort	Registry	61.9±14.1	65.50%	Develop a model without validation
Yu et al (2016)	USA	2008-2013	1	79	13		Retrospective cohort	EMR data	60(interquartile range: 51-70)	65.80%	External validation
Feng et al (2017)	China	2010.02-2014.12	1	136	39		Prospective cohort	EMR data	53.7±10.3	56.60%	Prediction performance of prognostic factors
Ghoreishi et al (2018)	USA	2002.01-2015.12	1	269		43	Retrospective cohort	EMR data	59±14	67.00%	Develop a model without validation
Zhang et al (2016)	China	2014.01-2015.06	1	67		26	Retrospective cohort	EMR data			Prediction performance of prognostic factors
Macrina et al (2010)	Italy	2002.01-late 2008	2	235		84	Prospective cohort	EMR data			Develop a model with internal validation
Macrina et al (2009)	Italy	2001.01-early 2008	2	208		53	Prospective cohort	EMR data	Survivors:61±12; Nonsurvivors: 66±10	64.00%	Develop a model with internal validation

Li et al (2019)	China	2007-2013.08	1	134	19	Prospective cohort	EMR data	Men: 50.59 ± 13.70, Women: 52.17 ± 11.55	67.3%	Prediction performance of prognostic factors
Centofanti et al (2006)	Multinational	1980-2004	Multicenter	616	154	Prospective cohort	Registry			Develop a model without validation
Santini et al (2007)		1979-2004		311	72	Retrospective cohort	EMR data	59.5 ± 13	72.00%	Develop a model without validation
Rampoldi et al (2007)	Multinational	1996-2003	18	682	163	Retrospective cohort	Registry	59.9 ± 13.8	70.30%	Develop a model without validation
Leontyev et al (2016)	Multinational	1996-2011	2	534	100	Prospective cohort	Registry	61 ± 14	63.70%	Develop a model without validation
Zhang et al (2019)	China	2013.11.01-2016.10.30	1	188	17	Prospective cohort	EMR data	57.7 ± 12.6	77.10%	Develop a model without validation
Zhang et al (2020)	China	2016.01-2019.06	1	186	40	Retrospective cohort	EMR data	50 ± 12	80.00%	Prediction performance of prognostic factors
Bedel et al (2019)	Finland	2013.01-2018.06	1	96	17	Retrospective cohort	EMR data	63.7±13.4	81.20%	Prediction performance of prognostic factors
Gong et al (2019)	China	2015.01-2017.05	1	583	70	Retrospective cohort	EMR data	48.05± 11.29		Prediction performance of prognostic factors

NLR: neutrophil lymphocyte ratio; hs-TnT: high-sensitivity cardiac troponin T; hs-CRP: high-sensitivity C-reactive protein; IRAD score: international registry of acute aortic dissection score; CRP: C-reactive protein; NT-proBNP: N-terminal pro-brain natriuretic peptide; BUN: blood urea nitrogen; TNC: Tenascin-C; EuroSCORE II: European System for Cardiac Operative Risk Evaluation.

Notes: The Liu et al (2018a) study and the Liu et al (2018b) study are the different prognostic models. Liu et al (2018a) study is for the relationship between fibrinogen and in-hospital mortality in patients with type A acute aortic dissection. Liu et al (2018b) study is for the relationship between blood urea nitrogen and in-hospital mortality of patients with acute aortic dissection.